National University of Computer & Emerging Sciences **CS2021: Object Oriented Data Structures** **(Fall 2026)**		**A-03** [Marks: 50+15]
Instructor: Shahid Qureshi	**Launch:** Oct 14, 2026	**Submit Date:** Mon, Oct 19, 2026
CLO-03 Apply OOP concepts of inheritance, overloaded operators and polymorphism to simplify problem solving		

Assignment Submission Guidelines:

- Late submissions of upto 12 hours are allowed but with a penalty of 20% on google classroom.
- **Add a header in your code file mentioning Assig#, Name, Roll# and a brief note about things which are working or are not working in your program**
- Name/rename your submission files as "Roll#-Assig#-filename.cpp", e.g., **"22i0899-A1-main.cpp".**
- Zero marks will be awarded in following cases
  - Code with build errors
  - Submitting after the late deadline
- Anyone can be asked to give demo of his submitted assignment.

Spell Checker using Hashing

You are provided a repository of English words in the form of "words.txt", which contains almost all the words used in the English language. You need to insert all these words from the repository into the hash table using chaining method.

After you have created the hash table, your program is ready to do the spell checking. Provide two options for getting text from the user for spell checking: 1) Getting input from command line, 2) Reading a "input.txt" file with any kind of text.

Provide appropriate menu for the above two options.
Whichever option is selected, your program should search for each word of the input text from the hash table to see if it's there or not. If the word is not found, then output such words on the screen.

To search a word from the hash table, it first needs to be converted to lowercase before the searching, because all the words in the hash table are lowercase.

All words not found will be displayed on the screen.

BONUS TASK
Rather than printing the just the words which were not found, reproduce the input text with missing words capitalized to make unknown words clearly visible.
For command line input, just display the reproduced text with missing words now capitalized, and for the text file input, create another file "output.txt" with same contents but with missing words now being capitalized.

Array Size:
Note that the repository contains ten thousand words, therefore select the array size accordingly. It is recommended to select a prime number between 9,000 and 10,000

Performance:
A good hash design will minimize collisions, and will hash the data to an empty slot. You need to work on the hash function and array size to minimize the collisions. **To check out, how good your hashing is, find and print the load factor for your hashing.**
Load Factor = Total hashed words / Total non-empty array slots

HashChain Class:
To implement hashing, you need to implement the following class design.

HashChain
Table: *list<string> size: int
HashChain(int): ~HashChain(); insert(string): void search(string): bool